Comenius University in Bratislava
Faculty of Mathematics, Physics and Informatics

New approaches to nowhere-zero flow problems

Diploma Thesis

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Study Programme:	Computer Science
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Department:	Department of Computer Science
Supervisor:	Mgr. Jozef Rajník, PhD.

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ZADANIE ZÁVEREČNEJ PRÁCE

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Názov: New approaches to nowhere-zero flow problems
Nové prístupy k problémom o nikde-nulových tokoch

Anotácia: Táto práca nadväzuje na výsledky bakalárskej práce rovnakého autora. Novo zavedený pojem viacrozmerných Manhattanských a Čebyševovských tokov ponecháva priestor pre ďalší výskum, ako napríklad hľadanie dolných odhadov pre tokové čísla. Jedným z hlavných výsledkov spomínanej bakalárskej práce je predstavenie hypotézy tvrdiacej, že každý bezmostový graf pripúšťa (1,2)-cirkulačnú dekompozíciu, t. j. 2-cirkuláciu a 4-cirkuláciu takú, že zakaždým keď je 2-cirkulácia nulová na nejakej hrane, tak 4-cirkulácia nemôže nadobúdať 0, +1 alebo -1. To ponúka bohatý priestor na výskum vrátane rôznych zovšeobecnení, v ktorých kladieme ďalšie požiadavky na tokové hodnoty.

Cieľ: 1. Dokázať netriviálne dolné odhady pre dvojrozmerné Čebyševovské tokové číslo grafu.
2. Preskúmať možné spôsoby, ako dokázať hypotézu, že každý bezmostový graf pripúšťa (1, 2)-cirkulačnú dekompozíciu, a iné súvisiace hypotézy. Dokázať túto hypotézu pre niektoré nekonečné triedy snarkov, prípadne pre niektoré snarky, ktoré majú ďaleko od toho, aby boli zafarbiteľné (napr. s nepárnosťou 2, indexom perfektného párenia 4, ...).
3. Preskúmať ďalšie zovšeobecnenia tokov potenciálne užitočné v kontexte vyššie uvedenej hypotézy.

Vedúci: Mgr. Jozef Rajník, PhD.
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Dátum zadania: 21.11.2024

Dátum schválenia: 05.12.2024

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THESIS ASSIGNMENT

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Title: New approaches to nowhere-zero flow problems

Annotation: This work builds on the results of the bachelor thesis of the same author. The newly introduced notion of multidimensional Manhattan and Chebyshev flows still leaves some possibilities for further research like finding lower bounds on flow numbers. One of the main results of the mentioned bachelor thesis is the introduction of the conjecture asserting that each bridgeless graph admits a (1,2)-circulation decomposition, that is a 2-circulation and a 4-circulation such that whenever the 2-circulation is zero on any edge, the 4-circulations can not attain 0, +1 or -1. This offers wide possibilities of exploration including various generalisations where further requirements are posed on the flow values.

Aim:
1. Prove nontrivial lower bounds on 2-dimensional Chebyshev flow number of a graph.
2. Explore possible ways of proving the conjecture that each bridgeless graph admits a (1, 2)-circulation decomposition, and other related conjectures. Prove this conjecture for some infinite families of snarks, eventually for some snarks that are far from being colourable (e.g. with oddness 2, perfect matching index 4, …)
3. Research other generalisation of flows potentially useful in the context of the abovementioned conjecture.

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Abstrakt

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Abstract

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Introduction

Chapter 1

Preliminaries

The class of type II cubic graphs is widely used, so its members have special name.

Definition. A *snark* is a bridgeless cubic graph, which has girth at least 5, is cyclically 4-edge connected and is not 3-edge-colourable.

Chapter 2

Cubic graphs generation

Chapter 3

Flows

In this chapter, we aim to introduce all concepts and current knowledge about flows and related graph theory terms. Mainly, we explain Chebyshev nowhere-zero flows, which have arisen in the recent preprint coauthored by the author [2]. Moreover, we present a concept of two smaller flows as a sufficient condition for a graph to have a larger nowhere-zero flow and some Chebyshev flow, also stated in that paper.

3.1 Integral nowhere-zero flows

In this thesis, we consider only unoriented simple connected graphs $G = (V, E)$. Then we can replace each unoriented edge uv with two oriented edges $\overrightarrow{uv}$, $\overrightarrow{vu}$, called *darts*. By a *dart-set* $D(S)$ of $S \subseteq V$ we mean the set of all darts with initial vertex in the set S.

Definition. Let $G = (V, E)$ be a graph and $k \geq 2$ be an integer. A *k-flow* on G is an assignment $\varphi\colon D(V) \to \{-(k-1), -(k-2), \ldots, k-2, k-1\}$ of flow values to the darts satisfying $\varphi(\overrightarrow{uv}) = -\varphi(\overrightarrow{vu})$ for any dart $\overrightarrow{uv} \in D(V)$ and

$$\sum_{\vec{d} \in D(\{v\})} \varphi\left(\vec{d}\right) = 0 \tag{3.1}$$

for each vertex v of G. The equality (3.1) is often referred to as the *flow conservation constraint*. Moreover, a *nowhere-zero k-flow* (or NZ k-flow) is a k-flow such that no flow value is zero, i.e. $\varphi\left(\vec{d}\right) \neq 0$ for any $\vec{d} \in D(V)$.

We also establish a notation $\varphi(e)$ for $e = uv \in E$, where $\varphi(e) = \max\{\varphi(\overrightarrow{uv}), \varphi(\overrightarrow{vu})\}$.

The flow conservation constraint (3.1) is generalisable to any subset of vertices. Hence, having a graph G with a bridge e, the interpretation of the flow conservation constraint for one of the components of $G - e$ results in the following observation.

Claim. If $e \in E$ is a bridge in G, then $\varphi(e) = 0$. Moreover, G allows a NZ k-flow for no natural k.

Therefore, nowhere-zero flows are examined only on bridgeless graphs.

It is not hard to see that a graph with an NZ k-flow also admits an NZ K-flow for any $K > k$. Hence it makes sense to seek for the smallest k such that a graph admits an NZ k-flow.

Definition. Let G be a bridgeless graph. A *flow number of* G is

$$\Phi(G) := \min\{k \mid \exists \text{NZ } k\text{-flow on } G\}.$$

However, it turns out that this rate of complexity achieves only few values.

Theorem 3.1 (6-flow theorem)**.** [5, p. 133] *There exists an NZ* 6*-flow on any bridgeless graph.*

Seymour constructed this NZ 6-flow from two simpler flows from the fact, that if $a \in \{0, \pm 1\}$, $b \in \{0, \pm 1, \pm 2\}$ and not both a, b are zero, then $3a + b \in \{\pm 1, \pm 2, \pm 3, \pm 4, \pm 5\}$.

Lemma 3.2. [5, p. 132] *For any bridgeless graph, there exist a* 2*-flow* φ_2 *and a* 3*-flow* φ_3 *such that for each edge* e*, at least one value from* $\varphi_2(e), \varphi_3(e)$ *is non-zero.*

On the other hand, there is no known graph with $\Phi(G) = 6$, all known graphs have $\Phi(G) \leq 5$. Therefore, the conjecture about the NZ 5-flow by Tutte still holds unresolved.

Conjecture 3.3 (5-flow conjecture)**.** [7, p. 83] *There exists an NZ* 5*-flow on any bridgeless graph.*

As for many problems in the graph theory, it is sufficient to explore the problem only on the smaller subset of graphs. There exists a folklore reduction of nowhere-zero flows to 3-regular (or cubic) graphs.

Proposition 3.4. *There exists an NZ* k*-flow on any bridgeless graph if and only if there exists an NZ* k*-flow on any cubic bridgeless graph.*

Hence, by considering only cubic graphs, we still solve the universal problem. Moreover, in terms of finding a counterexample to Tutte's 5-flow conjecture, we can also avoid 3-edge-colourable ones and limit research to the cubic graphs, that are not 3-edge-colourable, as a corollary of the next proposition.

Proposition 3.5. [1, pp. 160, 161] *A cubic graph has an NZ* 4*-flow if and only if it is* 3*-edge-colourable.*

3.2 Chebyshev flows and flow-pairs

There were some attempts to generalise nowhere-zero flows to real numbers and more dimensions. By generalising integer nowhere-zero flows to real numbers, researchers found out, that restricting only zero flow value is not sufficient, since any flow consisting of non-zero flow values can be made "smaller" and then, the notion of flow number disappears. Therefore, we need to determine a larger interval of real values forbidden as flow values, and hence, one-dimensional real nowhere-zero flows are allowed to use flow values from the interval $[1, r-1]$. The first attempts of generalisation real flows to more dimensions [4] tried to pick only vectors with the Euclidean norm from the mentioned interval $[1, r-1]$. It worked, but it did not have such nice properties, as we would expect. However, changing a norm turned out to be a good idea [2], so we will consider only Chebyshev multidimensional nowhere-zero flows.

Definition. Let $G = (V, E)$ be a bridgeless graph, d be a positive integer and $r \geq 2$ be a real constant. A *d-dimensional Chebyshev nowhere-zero r-flow* (or dD ChNZ r-flow) satisfies the conservation constraint (3.1) and the condition $1 \leq \|\varphi(e)\|_\infty \leq r-1$ for each edge $e \in E$.

The flow number can be defined in a similar way to integral nowhere-zero flows. The only problem is, that the set of real numbers does not need to contain a minimal element, so formally correct definition would use infimum instead of minimum. However, the infimum of the flow number candidates is always feasible [4, p. 101], so we can define the flow number straightforward with minimum.

Definition. Let G be a bridgeless graph and d be a positive integer. A *d-dimensional Chebyshev flow number* of G is

$$\Phi_d^\infty(G) := \min\{r \mid \exists d\text{D ChNZ } r\text{-flow on } G\}.$$

It has been proved [2, p. 4] that Chebyshev flow number is always rational. Moreover, the authors found some normal form of Chebyshev flows, as stated in the following proposition. Since we need the explicit construction in the latter proofs, we will give a sketch of this proof.

Proposition 3.6. [2, p. 6] *Let G be a graph with $\Phi_d^\infty(G) = p/q$ for some positive integers p, q. Then, there exists a dD ChNZ p/q-flow such that every coordinate of every flow value is an integral multiple of $1/q$.*

Sketch of the proof. If there is a dart with flow value, whose i^{th} coordinate is not an integral multiple of $1/q$, we can find an oriented cycle $\vec{\mathcal{C}}$ containing that dart with a property that no dart of this cycle has flow value with i^{th}, which is an integral multiple

of $1/q$. Otherwise, in some vertex, all but one dart have i^{th} coordinate of flow value, which is a multiple of $1/q$, and hence the sum of values on all darts (which is zero by the flow conservation constraint) is not a multiple of $1/q$, a contradiction. Now consider for each $\vec{d} \in \vec{\mathcal{C}}$ the maximal possible adjustment of $\varphi\left(\vec{d}\right)_i$ such that its absolute value remains in the range $[1, p/q - 1]$. Formally,

$$\delta\left(\vec{d}\right) = \begin{cases} p/q - 1 - \varphi\left(\vec{d}\right)_i & \text{if } \varphi\left(\vec{d}\right)_i > 0, \\ -1 - \varphi\left(\vec{d}\right)_i & \text{if } \varphi\left(\vec{d}\right)_i < 0. \end{cases}$$

Clearly $\delta\left(\vec{d}\right) > 0$ for any dart on cycle $\vec{\mathcal{C}}$, since both $p/q - 1, -1$ are integral multiples of $1/q$. Thus, by sending $\delta_{\min} = \min_{\vec{d} \in \vec{\mathcal{C}}} \delta\left(\vec{d}\right)$ along i^{th} coordinate, the number of conflicting coordinates decreases. By repeating this process, we get to a desired normal form. □

Remark. Note that no coordinate exceeds the nearest multiples of $1/q$ during the whole process. As a consequence, when we have a circular NZ r-flow, we can construct an integral NZ $\lceil r \rceil$-flow, where each flow value x is adjusted to either $\lfloor x \rfloor$ or $\lceil x \rceil$.

Now we recall the Seymour's flows from Lemma 3.2 – a 2-flow and a 3-flow with a property, that no edge has both flow values equal to zero. By simply using the value of the 2-flow as the first coordinate and the value of the 3-flow as the second coordinate, we get a Chebyshev nowhere-zero flow. Moreover, this flow gives us a nice bound on the Chebyshev flow number.

Proposition 3.7. [2, p. 8] *Each bridgeless graph has a 2D ChNZ* 3*-flow and hence,* $\Phi_2^\infty(G) \leq 3$.

Again, no graph with $\Phi_2^\infty(G) = 3$ is known. Futhermore, there is no known graph with $\Phi_2^\infty(G) > 5/2$. Therefore, the next conjecture is interesting to examine.

Conjecture 3.8. [2, p. 22] *Each bridgeless graph has a 2D ChNZ* 5/2*-flow and hence,* $\Phi_2^\infty(G) \leq 5/2$.

And similarly as for integral flows, the problem can be reduced to cubic graphs and then to snarks, since 3-edge-colourable cubic graphs have small flow number.

Theorem 3.9. [2, p. 14] *A cubic graph has a 2D ChNZ* 2*-flow if and only if it is* 3*-edge-colourable.*

As an NZ 6-flow and a 2D ChNZ 3-flow can be constructed from two simpler flows, something similar can be made for smaller flow numbers [2, p. 19]. We start by defining this structure called *flow-pair* and then we show that it satisfies desired properties.

Definition. Let $p \leq q$ be the positive integers. A *p/q-flow-pair* of a graph G is a 2-flow φ_2 and a $(p+q+1)$-flow φ_{p+q+1} such that whenever $\varphi_2(e)$ is zero, the value $\varphi_{p+q+1}(e)$ is at least q.

Remark. The structure in the Lemma 3.2 is a 1/1-flow-pair.

One possible ambiguity of this definition is, that since p/q is written as a fraction, the 2/6-flow-pair can be mixed up with the 1/3-flow-pair. However, their existence is equivalent (by a similar argument to Proposition 3.6), so we can WLOG assume that p, q are relatively prime.

Lemma 3.10. [2, p. 19] *A bridgeless graph G with a dp/dq-flow-pair also has a p/q-flow-pair and vice versa.*

We mentioned, that the p/q-flow-pair is a sufficient condition to the existence of some nowhere-zero flows. Those flows are somehow related with the value p/q and that is the reason to write it as a fraction in the definition of the flow-pair.

Proposition 3.11. [2, pp. 19, 20] *Consider a bridgeless graph G with a p/q-flow-pair. Then, there are a 2D ChNZ $(2+p/q)$-flow and a circular NZ $(4+2p/q)$-flow on G.*

Note that if there is a graph, which is a counterexample to Tutte's 5-flow conjecture, it should not have the 1/2-flow-pair. Hence it would be interesting to classify graphs with 1/2-flow-pair. By a computer, it was checked that this flow-pair exists for snarks up to 34 vertices, so it looks like the 1/2-flow-pair exists always.

Conjecture 3.12. [2, p. 22] *Each bridgeless graph has a 1/2-flow-pair.*

In general, edges with non-zero value of φ_2 form a set of cycles in G. In cubic graphs, these cycles should be clearly vertex-disjoint. Moreover, if considering other than trivial 1/1-flow-pair, these cycles together cover all vertices.

Lemma 3.13. [2, p. 21] *Let $p < q$ be positive integers. Consider a cubic graph G with a p/q-flow-pair $(\varphi_2, \varphi_{p+q+1})$. Then the set of edges e with a property $\varphi_2(e) \neq 0$ is a 2-factor of G.*

Chapter 4

Cubic multipoles generator

In this chapter, we introduce multipoles as elementary building blocks of graphs. Then we adjust known algorithms for generation of cubic graphs having given girth [**generation˙large˙girth**, **generation˙moore**] to obtain a generator of cubic multipoles with given girth.

Definition. A *multipole* is a graph where any edge can have 0, 1 or 2 incident endvertices. Edges with 1 incident endvertex are then called *dangling* and edges with no incident endvertices are called *isolated*.

First we modify an algorithm `snarkhunter` [**generation˙large˙girth**].

Definition. A *prime pole* is a multipole which can not be reduced to a smaller one by edge reduction.

Graph K_4^- is a K_4 after deleting any of its edges (so it looks like a diamond).

Lemma 4.1. *A multipole is a prime pole if and only if any of its edges is isolated, dangling, a bridge or has an endvertex in* K_4^-.

Chapter 5

On 1/2-flow-pairs

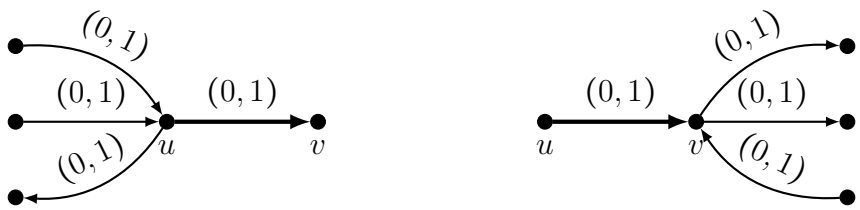
Theorem 5.1 (Menger theorem for oriented graphs)**.** ***TODO*** *https://doi.org/10.1112/jlms/s1-38.1.148*

Lemma 5.2. *Let G be a cyclically 4-edge connected graph with minimum degree at least 3. Consider a pair of vertices $u, v \in V(G)$ and a graph G', which arises from G by adding the edge uv. If G' has a Chebyshev NZ 2-flow φ satisfying*

(a) flow values are integral pairs – $\mathrm{Im}(\varphi) \subseteq \mathbb{Z} \times \mathbb{Z}$,

(b) $\varphi(\overrightarrow{uv}) = (0, 1)$, *and*

(c) these are not the flow configurations around u and v,



then G has a 1/2-flow-pair.

Proof. Multiply the second coordinate of φ by two to obtain $\varphi' = (\varphi'_1, \varphi'_2)$. Observe that the flow values of φ' are feasible for a 1/2-flow-pair and $\varphi'(\overrightarrow{uv}) = (0, 2)$. Since we are constructing a 1/2-flow-pair on $G = G' - uv$, the rest of the proof consists of finding two uv-paths, where we can send the flow $(0, 1)$ and get rid of the edge e.

Let H be an oriented graph such that $V(H) = V(G')$ and

$$E(H) = \{\overrightarrow{xy} \mid xy \in E(G') \wedge (\varphi_1(\overrightarrow{xy}) \neq 0 \vee \varphi_2(\overrightarrow{xy}) = 2)\}.$$

One can easily check that this corresponds to set of orientations, in which we can adjust the flow φ' by sending a value $(0, 1)$ and the flow values remain feasible. However, some

edges in G' induce two darts in H. But if we take both of these and send $(0,1)$ through each of them, the resulting flow value on the edges is the same as before the process. Hence, if we find two dart-disjoint uv-paths in $H - \overrightarrow{uv}$, together with edge uv they form two cycles. After sending $(0,1)$ along both of these cycles, the flow value at the edge uv becomes zero and all other flow values remain feasible, and therefore we obtain a 1/2-flow-pair of G.

From the oriented version of Menger's Theorem 5.1, there exist two dart-disjoint $\overrightarrow{uv}$-paths in $H - \overrightarrow{uv}$ if and only if each uv-dart-separator contains at least 2 darts from the component containing u to the component containing v. So it remains to prove this condition.

Consider any uv-dart-separator γ. Denote the darts from the component containing u to the component containing v as γ^+.

- If there is an edge having flow value with non-zero first coordinate in γ, there must be at least one more due to flow conservation constraint. Both of them are in H in both orientations.

- If $|\gamma| \geq 5$, the number of incoming $(0,2)$ edges must be equal to the number of outgoing $(0,2)$ edges. Therefore $|\gamma| \geq 6$, $|\gamma^+| \geq 3$, so there are at least two darts different from $\overrightarrow{uv}$.

- If $|\gamma| \leq 4$, the size of cut γ' in $H - \overrightarrow{uv}$ is equal to 3. Since the underlying graph G is cyclically 4-edge connected, at least one of the components must be acyclic mutlipole. Acyclic multipole with vertices of degrees $d_1, \ldots, d_k$ has $\sum_{i=1}^{k} d_i - k$ hanging edges. Together with the fact that $d_i \geq 3$, it has at least $2k$ hanging edges. With the fact that $|\gamma'| = 3$, we get $k = 1$. This leads to two forbidden configurations – cut around u and cut around v.

□

Corollary 5.2.1. Each snark of oddness 2 has a 1/2-flow-pair.

Theorem 5.3. *Let G be a cubic graph with 1/2-flow-pair. Then G has a rich nowhere-zero 7-flow.*

Proof. Denote the 1/2-flow-pair of G as $\varphi = (\varphi_1, \varphi_2)$. First, we construct a circular nowhere-zero 7-flow φ'. Then, we will adjust non-integral values of φ' to φ'', which will be the sought rich nowhere-zero 7-flow.

Let $\varphi'\left(\overrightarrow{d}\right) = 9\varphi_1\left(\overrightarrow{d}\right)/2 + \varphi_2\left(\overrightarrow{d}\right)/2$. We see that

- if $\varphi_1\left(\overrightarrow{d}\right) = -1$, then $\varphi'\left(\overrightarrow{d}\right) \in \langle -9/2 - 3/2, -9/2 + 3/2 \rangle$,

- if $\varphi_1\left(\overrightarrow{d}\right) = 0$, then $\varphi'\left(\overrightarrow{d}\right) \in \{-3/2, -2/2, 2/2, 3/2\}$, and

- if $\varphi_1\left(\overrightarrow{d}\right) = 1$, then $\varphi'\left(\overrightarrow{d}\right) \in \langle 9/2 - 3/2, 9/2 + 3/2\rangle$.

This shows that φ' is a circular nowhere-zero 7-flow.

By algorithm given in the proof of Proposition 3.6, we can construct an integral nowhere-zero 7-flow φ''. Remind that following mentioned algorithm, no flow value exceeds the nearest integers. Hence,

- if $\varphi_1\left(\overrightarrow{d}\right) = -1$, then $\varphi''\left(\overrightarrow{d}\right) \in \langle -6, -3\rangle$,
- if $\varphi_1\left(\overrightarrow{d}\right) = 0$, then $\varphi''\left(\overrightarrow{d}\right) \in \{-2, -1, 1, 2\}$, and
- if $\varphi_1\left(\overrightarrow{d}\right) = 1$, then $\varphi''\left(\overrightarrow{d}\right) \in \langle 3, 6\rangle$.

However, this is the key observation to prove that φ'' is rich. For the sake of contradiction, assume a vertex v in which flow values of φ'' in two outgoing darts are the same, denote it by x. Hence, the third flow value is $2x$, which is in absolute value greater. Hence, the two equal flow values must have the smallest absolute value. From Lemma 3.13, each vertex is surrounded by one dart with $\varphi_1\left(\overrightarrow{d}\right) = 0$ and two darts with $\varphi_1\left(\overrightarrow{d}\right) = 1$. But absolute values of $\varphi''\left(\overrightarrow{d}\right)$ on darts with $\varphi_1\left(\overrightarrow{d}\right) = 0$ are strictly smaller than absolute values of $\varphi''\left(\overrightarrow{d}\right)$ on darts with $\varphi_1\left(\overrightarrow{d}\right) \neq 0$. Therefore, there cannot be two darts with the same smallest absolute value, a contradiction. In final, we get that φ'' is a rich NZ 7-flow.

□

Chapter 6

Chebyshev flow number of snarks

6.1 Chebyshev flow number of generalised Blanuša snarks

6.2 General lower bound for Chebyshev flow number

In one-dimensional flows, there has been proved a lower bound on the flow number of a snark

$$\Phi_1(G) \geq 4 + 1 \Big/ \left\lceil \frac{n-4}{8} \right\rceil$$

depending on the number n of its vertices [3, p. 14]. Moreover, there is also a meaningful lower bound on the two-dimensional Chebyshev flow number of the Petersen graph P [6, p. 99]. It states that $\Phi_2^\infty(P) \geq 5/2$. We provide generalisation of this bound for any snark. Moreover, the bounding number is roughly a half of the bounding number in one dimension.

Proposition 6.1. *Let G denote a snark of order n. Then $\Phi_2^\infty(G) \geq 2 + 1/\xi$, where $\xi = \left\lfloor \frac{n-2}{4} \right\rfloor$.*

Proof. Assume by contradiction that there exists a 2-dimensional flow φ of G such that $\varphi = (\varphi_1(e), \varphi_2(e))$ for each edge $e \in E(G)$, with $\|\varphi(e)\|_\infty \geq 1$ and $\varphi_i(e) \in (-1 - 1/\xi, 1 + 1/\xi)$ for $i = 1, 2$.

We say an edge $e \in E(G)$ to be *large* with respect to φ_i if $|\varphi_i(e)| \in [1, 1 + 1/\xi)$, *small* otherwise. Observe that an edge e can be large with respect to both φ_1 and φ_2, but it cannot be small with respect to both φ_1 and φ_2, for otherwise $\|\varphi(e)\|_\infty < 1$.

Denote by S_i the subgraph of G induced by the small edges with respect to φ_i and by L_i the one induced by the large edges with respect to φ_i, $i = 1, 2$. By previous

observation at least one between S_1 and S_2 has at most $\lfloor |E(G)|/2 \rfloor = \lfloor \frac{3n}{4} \rfloor$ edges, say S_1.

Claim. S_i is a spanning subgraph of G and $\Delta(S_i) \leq 2$, for $i = 1, 2$.

Proof. is this needed?? Observe that $\Delta(L_i) \leq 2$, because the sum of three real numbers all with absolute value in the interval $[1, 1 + 1/\xi)$ cannot give 0 as a result, making the Kirkoff's law impossible to be satisfied by φ around a vertex of G. Hence S_i is spanning, for otherwise $\Delta(L_i) = 3$ and $\Delta(S_i) \leq 2$, for otherwise $\Delta(L_{3-i}) = 3$. □

Claim. If $C \subseteq E(L_i)$ is an odd edge-cut of G for $i = 1, 2$, then $|C| \geq 2\xi + 3$.

Proof. Consider an odd edge-cut C of G containing $2k + 1$ large edges, separating components G_1, G_2. WLOG assume $\varphi_i(e) \geq 0$ for every edge e of C and more edges are directed towards G_2. Then, there are at least $k+1$ edges towards G_2, resulting in total inflow at least $(k+1) \cdot 1$. Analogously, the total outflow is strictly less than $k \cdot (1+1/\xi)$. Together with the Kirkoff's law, this leads to $k + 1 < k \cdot (1 + 1/\xi)$, which is equivalent to $k > \xi$. □

Claim. A path of length $2k$ cannot be a connected component of S_i, for $k = 1, 2, \ldots, \xi - 1$ and $i = 1, 2$.

Proof. For the sake of contradiction assume there is a path of a length $2k$, $k < \xi$ in S_i. Then the edges adjacent to this path are in L_i and they form an odd edge-cut of G, containing at most $2k + 3 < 2\xi + 3$ edges, which is in contradiction with the Claim 2. □

Claim. A cycle of length $2k+1$ cannot be a connected component of S_i, for $k = 1, 2, \ldots, \xi$ and $i = 1, 2$.

Proof. For the sake of contradiction assume there is a cycle of a length $2k + 1$, $k < \xi + 1$ in S_i. Then the edges adjacent to this cycle are in L_i and they form an odd edge-cut of G, containing at most $2k + 1 < 2\xi + 3$ edges, which is in contradiction with the Claim 2. □

Claim. $E(S_i)$ cannot contain a perfect matching of G, for $i = 1, 2$.

Proof. For the sake of contradiction assume that $E(S_i)$ contains a perfect matching M of G. Then also $E(L_{3-i})$ contains a perfect matching M of G. Next, $F = E(G) \setminus M$ is a 2-factor of G. Note that F must contain an odd cycle of length $2k + 1 \leq \frac{n}{2}$ explain more – here's implicitly used that G is snark. This is equivalent with $k \leq \xi$. Then the edges adjacent to this cycle are in L_{3-i} and they form an odd edge-cut of G, containing at most $2k + 1 \leq 2\xi + 1$ edges, which is in contradiction with the Claim 2. □

Note that by the Claim 3, each even path in S_1 contains at least 2ξ edges. Similarly by the Claim 4, each odd cycle in S_1 contains at least $2\xi+3$ edges. Let *odd components* denote odd cycles and even paths. Since $4\xi = 4\lfloor\frac{n-2}{4}\rfloor > \lfloor\frac{3n}{4}\rfloor \geq E(S_1)$ obviously holds for any $n \geq 10$, S_1 may contain at most one odd component. On the other hand, S_1 contains even number of odd componentss. As a result, S_1 contains only odd paths and even cycles, but then also a perfect matching of G, which is in contradiction with the Claim 5. □

So now, not only for the Petersen graph P, but also for both Blanuša snarks S_1, S_2 this bound is tight, since there are a ChNZ 5/2-flow on P and ChNZ 9/4-flows on S_1, S_2. Hence, $\Phi_2^\infty(P) = 5/2$ and $\Phi_2^\infty(S_1) = \Phi_2^\infty(S_2) = 9/4$.

6.3 Three-dimensional Manhattan flow number

Proposition 6.2. *For any bridgeless graph G it holds $\Phi_3^1(G) \leq \frac{5}{2}$.*

Proof. Define the following vectors in $\mathbb{R}^3$:

$$q_1 = \left(\frac{3}{8}, \frac{3}{8}, -\frac{1}{4}\right),$$
$$q_2 = \left(\frac{3}{8}, -\frac{1}{4}, \frac{3}{8}\right),$$
$$q_3 = \left(-\frac{1}{4}, \frac{3}{8}, \frac{3}{8}\right).$$

Note that since the vectors are cyclic, it suffices to find norms of $q_1, q_1 + q_2, q_1 - q_2, q_1 + q_2 + q_3$ and $q_1 + q_2 - q_3$. But

$$\|q_1\|_1 = 1,$$
$$\|q_1 + q_2\|_1 = 1,$$
$$\|q_1 - q_2\|_1 = \frac{5}{4},$$
$$\|q_1 + q_2 + q_3\|_1 = \frac{3}{2},$$
$$\|q_1 + q_2 - q_3\|_1 = \frac{3}{2}.$$

By the 8-flow theorem, the graph G can be covered by three cycles C_1, C_2, C_3. We orient each cycle arbitrarily and assign to each oriented cycle C_i the flow vector q_i. The total flow is then defined as the sum of these three cycle flows.

By construction and the norm evaluations above, for every edge $e \in E$, the resulting flow satisfies

$$\|\varphi(e)\|_1 \in \left\{1, \frac{5}{4}, \frac{3}{2}\right\}.$$

□

Conclusion

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